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SEAT NO. CT-28

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Computer Science)

SPRING SEMESTER EXAMINATIONS 2026

Batch 2023

Time : 3 Hours

Dated : 10-JUN-26
Max Marks : 60

Natural Language Processing - CT-485

Q1) (a) Explain Word Embeddings in detail and how they differ from the Bag of Words model. Also discuss various applications of word embeddings in Natural Language Processing.

(b) Compare Word2Vec, GloVe, BERT, and GPT.

[CLO-1: 12 Marks]

Q2) (a) Discuss problem of Zero Frequency in Naive Bayes and explain how Laplace Smoothing solve it.

(b) Compare the sigmoid, tanh, and ReLU activation functions in terms of their output behavior.

(c) An AI recruitment system rejects candidates unfairly because of biased training data.

- Explain the role of bias in this scenario.
- Suggest methods to improve fairness.

[CLO-1: 12 Marks]

Q3) Analyze the following example case related to the Bag-of-Words model.

- *NLP improves language understanding*
- *NLP models understand text data*

a) Create a Bag of Words (BoW) representation for these two sentences.
b) Calculate the TF (Term Frequency) for each word.
c) Calculate the IDF (Inverse Document Frequency) for each word.
d) Compute the TF-IDF values for each word.

[CLO-2: 12 Marks]

Q4) Analyze the following email dataset using Naive Bayes.

Email Text	Class
"Win money now"	Spam
"Cheap offer today"	Spam
"Project meeting attached"	Ham
"Schedule the project discussion"	Ham

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"Win project offer"

a) Build the vocabulary.
b) Calculate prior probabilities for Spam and Ham.
c) Calculate likelihood probabilities using Laplace smoothing.
d) Compute posterior probabilities.
e) Predict whether the email is Spam or Ham.

[CLO-2: 12 Marks]

Q5) Apply a bigram language model to the following corpus and answer the Questions given below.

Training data:

<s> machine learning improves </s>
<s> deep learning models </s>
<s> machine models learn </s>

1. Predict the most probable next word for:

(1) <s> machine . . .
(2) <s> deep learning . . .

2. Which of the following sentences is better, i.e., gets a higher probability with this model?

(1) <s> machine learning models </s>
(2) <s> deep learning improves </s>

[CLO-3: 12 Marks]